DO NOW — RECALL

# Waterfall, Agile and Extreme Programming

Name: _____ Class: _____ Date: _____

**1. Which OCR pseudocode function reads the next line of a text file?**

**2. What is a function parameter?**

**3. Why use boundary-value testing?**

**4. If requirements frequently change, what is the risk of freezing all requirements before implementation?**

**DO NOW — CHECK / OBJECTIVES**

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

readLine() on an opened file object.

### 2. Model answer

A named input received by a subroutine.

### 3. Model answer

Defects often occur at the limits of valid ranges.

### 4. Model answer

Large amounts of design/code may need reworking when changes emerge.

### Vocabulary — know these terms

Waterfall — Primarily sequential software-development life cycle.

Agile — Iterative approach using feedback and incremental releases.

XP — Extreme Programming, an agile methodology with disciplined engineering practices.

sprint — Short time-boxed development iteration.

test-driven development — Write an automated test before writing the production behaviour.

NEW LEARNING

# Read, annotate and explain

### Waterfall is sequential

Agree requirements, then design, implementation, testing and deployment. Revisiting an earlier stage can be costly.

### Agile delivers incrementally

Each short iteration produces working software and seeks feedback, so priorities can change between iterations.

### Extreme Programming (XP)

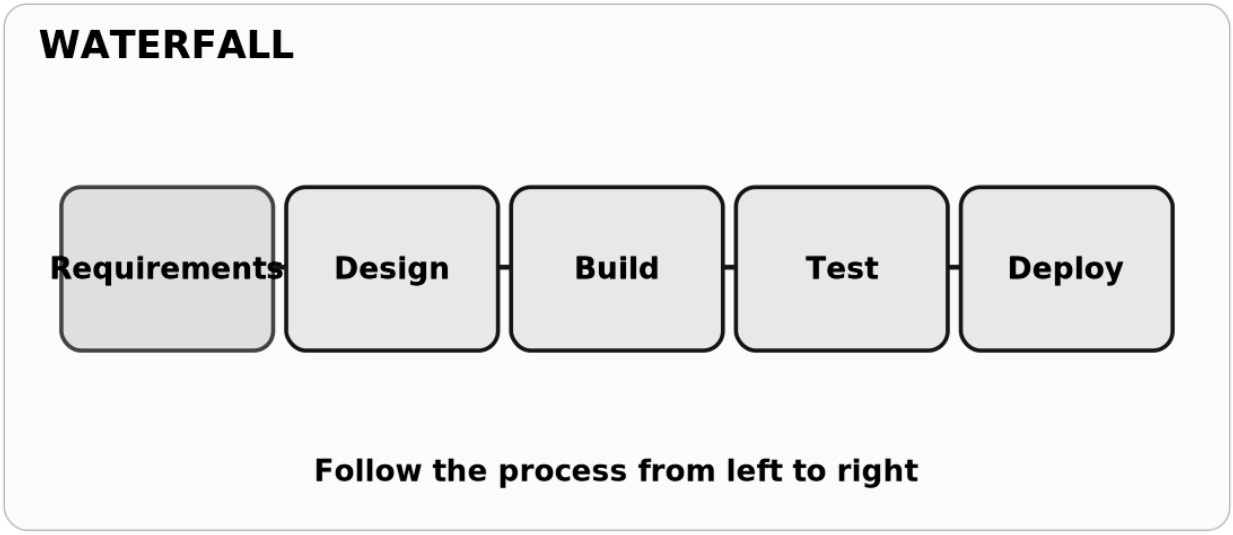
XP is an agile methodology emphasising practices such as test-driven development, pair programming, refactoring and continuous integration.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

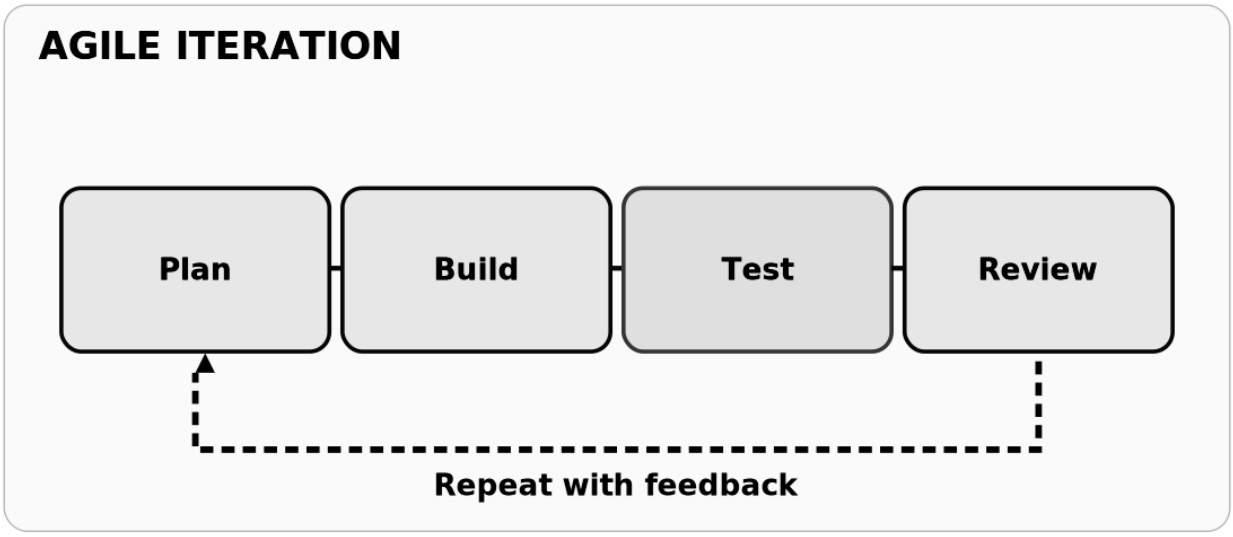
Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning




WATERFALL; Requirements to Design to Build to Test to Deploy

### New Learning

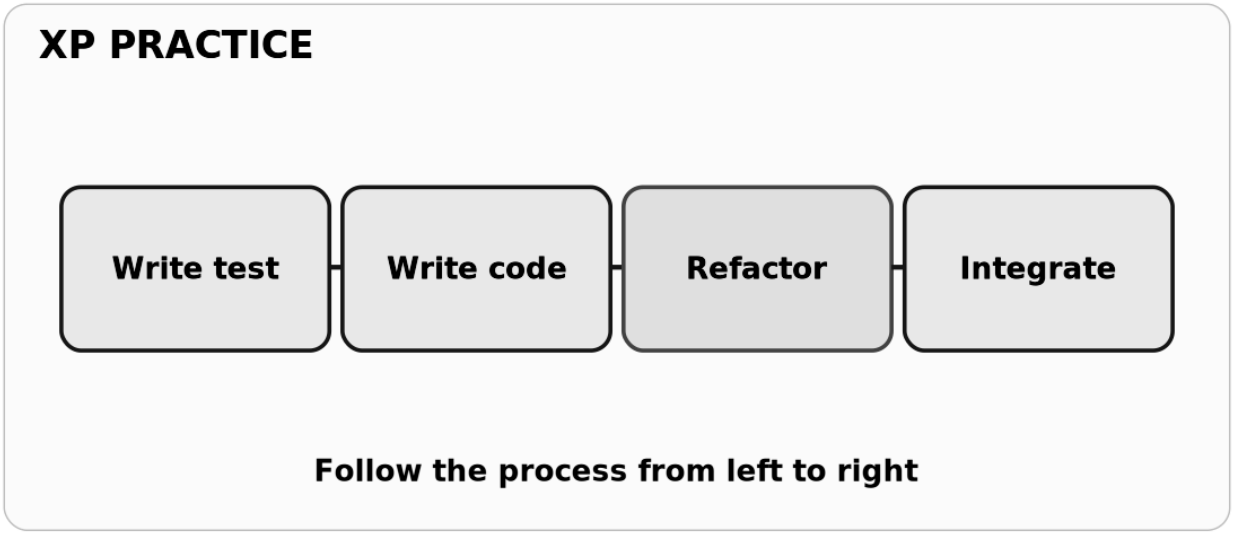



AGILE ITERATION; Plan to Build to Test to Review

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning




XP PRACTICE; Write test to Write code to Refactor to Integrate

I DO — WORKED EXAMPLE

# Follow the worked method

**A hospital requests software with a carefully agreed specification and formal approvals. Why might Waterfall suit it?**

### Predict first

Requirements and approval milestones can be documented before later stages proceed.

### Model and explain

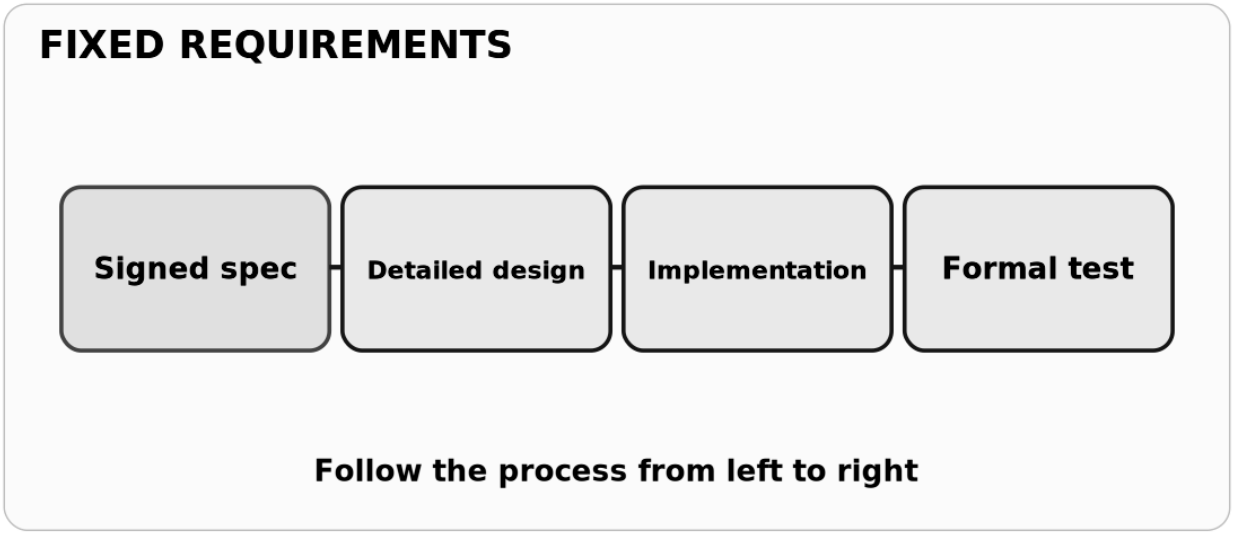
Waterfall offers clear documentation and control, but late changes can require expensive rework.

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### A hospital requests software with a carefully agreed specification and formal approvals. Why might Waterfall s

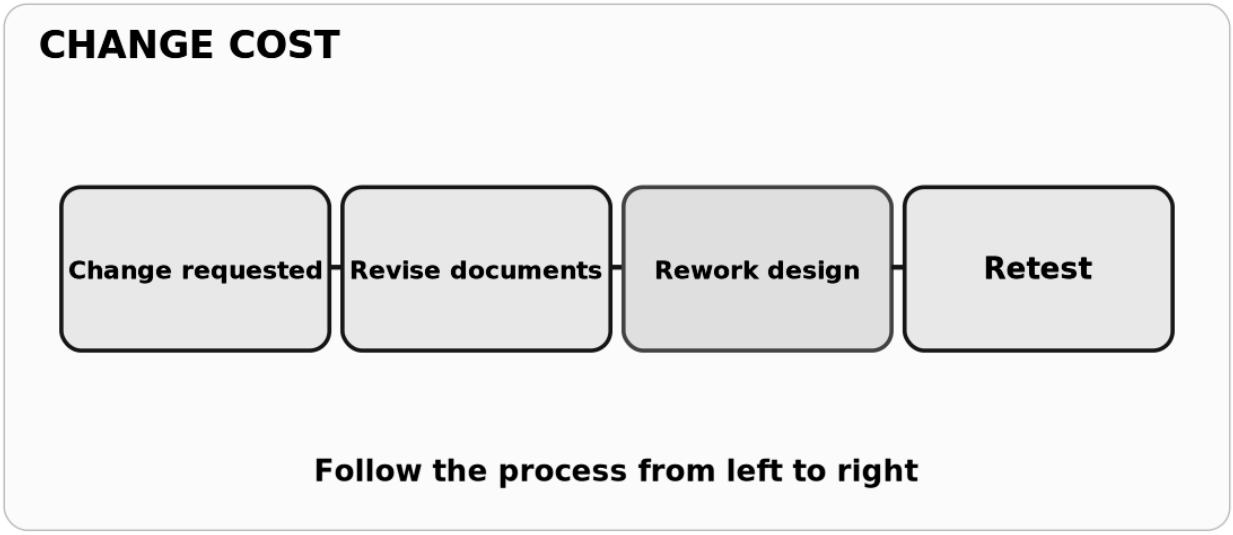
Requirements and approval milestones can be documented before later stages proceed.




FIXED REQUIREMENTS; Signed spec to Detailed design to Implementation to Formal test

### A hospital requests software with a carefully agreed specification and formal approvals. Why might Waterfall s

Requirements and approval milestones can be documented before later stages proceed.




CHANGE COST; Change requested to Revise documents to Rework design to Retest

WE DO — GUIDED PRACTICE

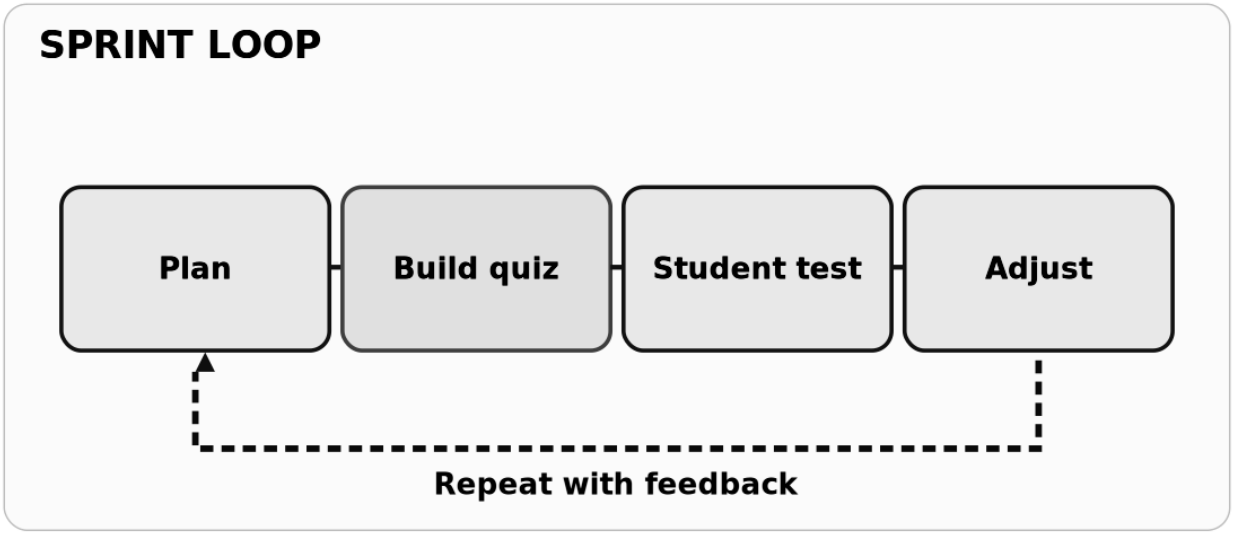
# Try → Discuss → Improve

# We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**A pupil revision app changes after each group of students tries it. Which approach fits and why?**

Changing needs and quick feedback favour Agile iterations over a rigid up-front plan.




SPRINT LOOP; Plan to Build quiz to Student test to Adjust

# WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

### Guided 1: Predict first Changing needs and quick feedback favour Agile iterations over a rigid up-front plan.

DISCUSS — Compare the sequence of steps or states. Identify the first difference in your methods and justify which step is correct.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

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YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

State a defining characteristic of Waterfall, Agile and XP.

### Question 2

A mobile app company receives new customer requirements each fortnight. Recommend a methodology and explain two advantages and one disadvantage.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Explain how test-driven development and continuous integration in XP could reduce defects, and why neither guarantees fault-free software.

**EXIT TICKET / REFLECTION**

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**Agile means software is never planned or tested.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

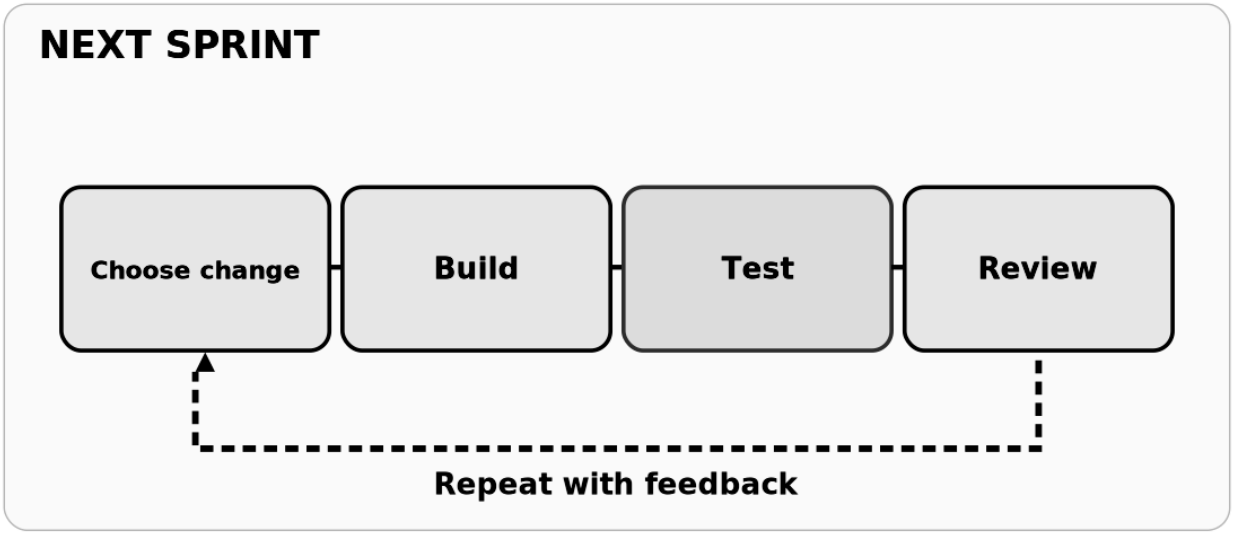
Next step: revisit the model or a relevant revision-guide section if needed.

# We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**A pupil revision app changes after each group of students tries it. Which approach fits and why?**

Changing needs and quick feedback favour Agile iterations over a rigid up-front plan.




NEXT SPRINT; Choose change to Build to Test to Review

# WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

### Guided 1 — original CHECK

Model and explain

A usable quiz is released first, then revision history and accessibility features improve in later sprints.

IMPROVE: Explain one correction to your own reasoning.